Submission Summary

Conference Name

Fourth International Conference on Data Science and Network Engineering

Track Name

Track 6: Computing Technologies

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66

Paper Title

PersistHotStuff: Designing a Practical Byzantine Fault-Tolerant Consensus

Abstract

Building a working Byzantine Fault-Tolerant (BFT) consensus system is harder than the theory suggests. HotStuff offers an elegant O(n) protocol with a clean 3-chain commit rule, yet every prototype we studied shares the same blind spots: no crash-safe storage, fixed validator sets, a toy state machine, and a pacemaker that stalls when clients go quiet. We set out to close these gaps. PersistHotStuff is our Rust-based framework built around four proposed extensions to HotStuff: a WAL-and-snapshot persistence layer, consensus-driven dynamic membership, a gRPC-backed pluggable state machine interface, and a pacemaker that injects dummy blocks to keep the 3-chain moving. We provide a TLA+ specification of the dynamic membership protocol, verified by TLC model checking across 7.3 million distinct states with zero violations of 11 safety invariants, and benchmark the system against analytical models of PBFT and Raft, confirming O(n) message complexity with 9× fewer messages than PBFT at n=13.

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Submission Files

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